

# Econometrically Disciplined Forecasting of Daily Equity Returns: A Comparative Study of Recurrent and Transformer Sequence Models on General Electric

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**Abstract**—This report analyzes the daily price behavior of General Electric (GE) from January 2020 through May 2026 and evaluates whether modern sequence models can forecast its next-day log returns. Texas Instruments (TXN) is used as a comparison security. Rather than regressing raw, non-stationary prices, all predictive work targets the daily log return, which an Augmented Dickey–Fuller (ADF) test confirms is stationary. We benchmark a Long Short-Term Memory (LSTM) network and a lightweight time-series Transformer encoder against two naive baselines—a zero-return predictor and a persistence (last-value) predictor—using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), the coefficient of determination ( $R^2$ ), and Theil’s  $U$ . Although both deep models register Theil’s  $U < 1$  against persistence, their negative  $R^2$  values indicate they do not extract genuine predictive signal from daily returns. The dominant empirical lesson is that daily equity returns are close to unforecastable, and that beating a deliberately weak baseline is not equivalent to demonstrating real skill.

**Index Terms**—financial time series, log returns, stationarity, LSTM, Transformer, persistence baseline, Theil’s  $U$

## I. INTRODUCTION

Equity price series are a classic testbed for sequence modeling, but they are also a classic trap. Raw closing prices are strongly trending and non-stationary, so a model can appear accurate simply by tracking the trend rather than by predicting anything new. The goal of this study is to apply machine learning to General Electric while avoiding two common methodological errors: forecasting raw prices instead of returns, and judging models against an artificially low bar.

General Electric is a useful subject because its 2020–2026 history is shaped by a genuine structural event—the breakup into GE Aerospace and GE Vernova—rather than by a smooth, featureless drift. Texas Instruments serves as a contrasting semiconductor name for descriptive comparison. The analysis proceeds from descriptive trend and volume statistics, through a stationarity diagnosis, to a head-to-head comparison of an LSTM and a Transformer encoder against two baselines.

## II. DATA AND METHODOLOGY

### A. Data

The primary GE series spans 2020-01-01 to 2026-05-26 (a long history), while the TXN comparison series spans 2024-01-01 to 2026-05-26. Data were obtained through the `yfinance` interface and stored as CSV files. Missing values introduced for the cleaning exercise were repaired with forward fill, which propagates the most recent *past* observation and therefore respects the arrow of time.

### B. Stationarity and Target Selection

We deliberately avoid regressing raw `Close` on same-day prices. The daily log return is defined as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \quad (1)$$

where  $P_t$  is the closing price on day  $t$ . An ADF test was applied to both the raw price and the log return to justify this choice.

### C. Models and Evaluation

Predictive features are lagged and causal: only information available strictly before day  $t$  is used to predict  $r_t$ . Inputs are arranged into sliding windows of 25 trading days, and the train/test split is sequential (80/20) with no shuffling, preserving temporal order. Two deep models are compared:

- an **LSTM** recurrent network, chosen over a vanilla RNN because gated memory mitigates the vanishing-gradient problem over a 25-day window; and
- a **time-series Transformer encoder** with sinusoidal positional encoding applied to projected continuous features (not NLP token embeddings).

Each is compared against a **zero-return baseline** (always predict  $r_t = 0$ ) and a **persistence baseline** (predict  $r_t = r_{t-1}$ ). Models are scored with MAE, RMSE,  $R^2$ , and Theil’s  $U$ ,

$$U = \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}, \quad (2)$$

where  $U < 1$  indicates the model beats the naive last-value forecast.

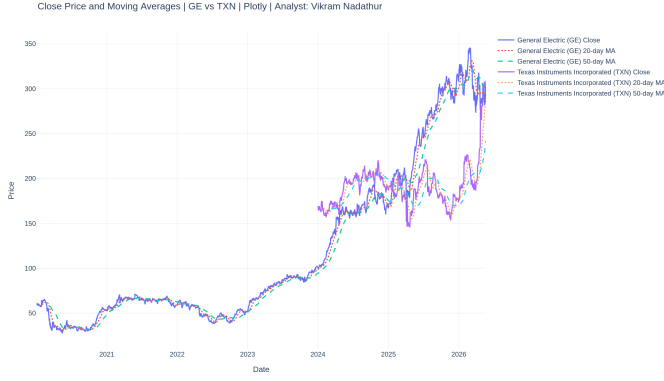


Fig. 1. Moving-average comparison for GE and TXN. The ordering of the 20/50/100-day averages reflects a durable uptrend in GE over the analysis window.

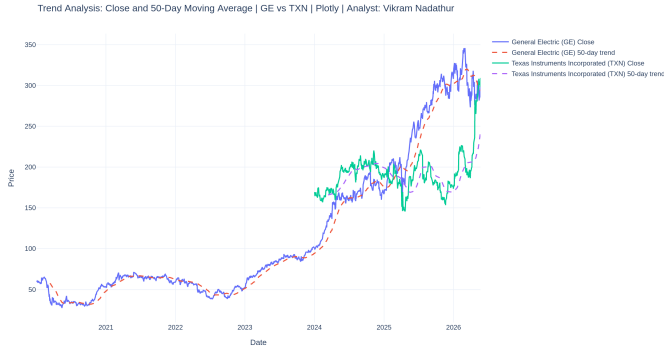


Fig. 2. 50-day rolling-mean trend for both securities. GE's trend is upward with normal short-term volatility rather than a smooth straight line.

### III. DESCRIPTIVE ANALYSIS

Over the full window, GE's closing price rose from approximately \$59.46 to \$302.84, a gain of about 409%. The minimum close of \$27.36 occurred on 2020-05-15, during the early-pandemic drawdown, and the maximum close of \$345.74 was reached on 2026-03-02. As shown in Fig. 1, the 20-, 50-, and 100-day moving averages stack in trend order—shorter averages above longer ones—for most of the later period, the canonical shape of a sustained uptrend. Brief pullbacks below the 50-day average appear in 2022 before the trend resumes (Fig. 2).

Volume analysis highlights episodic spikes well above each stock's median. GE's largest-volume sessions cluster in 2021–2022 (for example 2021-11-09 and 2022-04-26, each exceeding 40 million shares), consistent with restructuring announcements and earnings reactions. TXN's volume spikes fall in late 2025 through early 2026 around earnings dates.

### IV. STATIONARITY RESULTS

The ADF test makes the case for modeling returns rather than prices explicit. For the raw closing price the test statistic is +1.34 with  $p \approx 0.997$ , so the unit-root null cannot be rejected and any regression on the raw level risks being spurious. For the daily log return the statistic is  $-24.29$  with

TABLE I  
NEXT-DAY LOG-RETURN FORECASTING PERFORMANCE (GE)

| Model                | MAE    | RMSE   | $R^2$    | Theil's $U$ |
|----------------------|--------|--------|----------|-------------|
| Zero-Return Baseline | 0.0153 | 0.0215 | $-0.005$ | 0.713       |
| LSTM                 | 0.0153 | 0.0216 | $-0.015$ | 0.716       |
| Transformer          | 0.0242 | 0.0294 | $-0.887$ | 0.976       |
| Persistence Baseline | 0.0226 | 0.0301 | $-0.980$ | 1.000       |

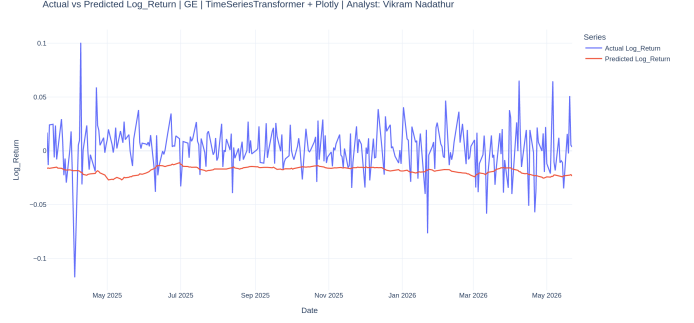


Fig. 3. Time-Series Transformer: actual vs. predicted daily log returns on the GE test set. Predictions remain near zero and do not track large realized moves.

$p \approx 0.000$ , strong evidence against a unit root and consistent with stationarity. This is precisely why the linear-regression, recurrent, and Transformer sections all target  $r_t$  rather than  $P_t$ .

### V. FORECASTING RESULTS

Table I reports the next-day log-return forecasting performance of all four approaches on the held-out test period.

Two observations stand out. First, on Theil's  $U$  every model except persistence scores below 1.0, so in a narrow sense the LSTM ( $U \approx 0.716$ ) and Transformer ( $U \approx 0.976$ ) both “beat” the naive last-value forecast—and so does the trivial zero-return baseline ( $U \approx 0.713$ ). Second, and more importantly, all models post negative or near-zero  $R^2$ , meaning none of them explains next-day return variance better than simply predicting the mean. The LSTM clearly outperforms the Transformer here, with lower MAE, lower RMSE, and a stronger Theil's  $U$ ; the Transformer's larger errors suggest it overfits or chases noise on this small, low-signal target. Fig. 3 shows the Transformer's predicted versus actual returns, where the predictions hug zero and fail to anticipate the sharp daily moves.

### VI. DISCUSSION

The results are a textbook illustration of why financial forecasting demands disciplined baselines. A model can satisfy Theil's  $U < 1$  and still be useless, because the persistence baseline for daily returns is a poor reference: tomorrow's return is rarely yesterday's return. The zero-return baseline—essentially the efficient-market null—is far harder to beat, and the fact that the LSTM only ties it (MAE 0.0153 for both) is the real story. Negative  $R^2$  across the board confirms that daily

GE returns behave close to a martingale difference: nearly unforecastable from their own recent history.

This does not invalidate the modeling pipeline; it validates it. By targeting a stationary series, using causal lagged features, splitting sequentially, and comparing against multiple baselines, the analysis avoids the false confidence that a raw-price model would have produced. The honest conclusion is negative, and a negative result obtained rigorously is more informative than an impressive-looking one obtained by leaking the trend.

## VII. CONCLUSION

Descriptively, General Electric exhibited a strong multi-year uptrend (about 409% appreciation) driven largely by its post-2020 recovery and restructuring. Predictively, neither an LSTM nor a Transformer demonstrated genuine skill at forecasting next-day log returns: both fell short of the efficient-market zero-return baseline on  $R^2$  despite nominally beating persistence on Theil's  $U$ . The study's contribution is methodological—showing how stationarity testing, causal feature construction, sequential splitting, and multi-baseline evaluation together prevent the spurious results that naive raw-price modeling invites. Future work could incorporate exogenous features (volatility regimes, volume, macro indicators) or shift the target to realized volatility, which is more persistent and therefore more plausibly forecastable than the return itself.

## NOTE

This report is an educational exercise and does not constitute investment advice. All metrics are taken directly from the author's executed analysis notebook for GE and TXN.